IILM UNIVERSITY, GREATER NOIDA | SUMMER INTERNSHIP PRESENTATION 2026

Python Developer Internship

URL Shortener Application (GUI + Persistent Storage)

Student Name: Prince Maurya
Roll No.: 25SCS1003001538 | **Section:** 2CSE7 (3rd Sem)
Programme: B.Tech Computer Science & Engineering (2025-2029)
Industry Partners: upSkill Campus | UniConverge Technologies (UCT) | AICTE
Certificate ID: USC838112TIA (12-08-2026 to 02-09-2026)

01. About the Student

Academic Details

Student Name: Prince Maurya
Roll Number: 25SCS1003001538
Department: CSE, IILM University
Batch: 2025-2029 (3rd Semester)
Section: 2CSE7

Internship Credentials

Domain: Python Programming
Role: Python Developer Intern
Partners: upSkill Campus + UCT + AICTE
Duration: 12-08-2026 to 02-09-2026
Cert ID: USC838112TIA

02. Internship Objectives

- **Core Python Mastery:** Strengthen fundamentals, logic building, data structures, and function definitions.
- **Event-Driven Programming:** Build an interactive Desktop GUI using Python's `tkinter` library.
- **Data Persistence:** Master local file I/O operations and JSON serialization using the `json` module.
- **Input Validation & Exception Handling:** Prevent crashes using `try-except` blocks and string checking.
- **Project Delivery:** Build, test, and host a complete URL Shortener project on GitHub.

03. Learning Progress & Milestones

4-WEEK STRUCTURED ROADMAP

Sprint	Focus Area	Key Deliverables
Week 01	Python Core & Logic	Selected URL Shortener project; developed initial CLI code using `string` & `random`.
Week 02	Data Persistence	Integrated `json` file storage (`url_database.json`) and added HTTP/HTTPS validation.
Week 03	GUI Development	Designed desktop window interface using `tkinter`; integrated OS clipboard copy functionality.
Week 04	Testing & Submission	Added `messagebox` error alerts, "Clear All" button, completed testing, pushed to GitHub.

04. Project Overview: URL Shortener

The Problem

Long, cumbersome web URLs are hard to share in documents, presentation slides, or emails. Public cloud solutions (Bitly) require paid tiers, track users, and require constant internet connectivity for local management.

The FixMate-Style Solution

A Python desktop app with a clean GUI that converts long links into unique 6-character short codes, backed by persistent local JSON storage for privacy, speed, and 100% offline usability.

05. Technology Stack

CORE TOOLS & MODULES

Python 3.x: Main language & backend logic

String & Random: 6-character code generation

Tkinter: GUI windows, inputs, buttons

OS Module: File existence verification

JSON Module: Persistent disk database storage

Git / GitHub: Version control & repo hosting

06. System Architecture

END-TO-END EXECUTION FLOW

[User Inputs URL in GUI] → [Protocol Validation] → [Collision-Free Code Generator] → [JSON Local File Save] → [Clipboard Copy / GUI Result]

- **Validation Layer:** Ensures URL starts with `http://` or `https://`.
- **Algorithmic Layer:** Creates randomized 6-character alphanumeric combinations.
- **Database Layer:** Synchronizes Python dictionaries directly with `url_database.json`.

07. Core Application Features

- **Instant URL Shortening:** Generates unique 6-character links instantly.
- **Collision Prevention:** Automated loop prevents duplication of existing short codes.
- **Original URL Retrieval:** Enter short code to fetch original long URL.
- **Clipboard Integration:** Single-click copy to system clipboard (`root.clipboard_append`).
- **Defensive Validation:** Pop-up alerts (`messagebox`) prevent empty/malformed inputs.

08. Testing & Verification

QUALITY ASSURANCE CHECKS

Test Scenario	Observed Behavior	Result
Empty Input Submitted	Warning messagebox popup shown; no crash	**PASS**
Invalid URL Format	Format warning popup shown	**PASS**
Clipboard Copy Clicked	Short link copied to OS memory successfully	**PASS**
Application Reboot	All shortened URLs persist in `url_database.json`	**PASS**

09. GitHub Submission

GitHub Repository Link

`https://github.com/Princemau25/url-shortener`

College Standard Naming Convention

Format: Session_Name_RollNo_Semester_Section
Details: 2025-2029_Prince_Maurya_25SCS1003001538_3rd_sem_2CSE7

10. Conclusion & Future Scope

- **Key Takeaways:** Bridged Python fundamentals with real-world software design, mastering Tkinter GUI building, JSON persistence, and GitHub documentation.
- **Future Enhancements:**
  - Migrate local JSON storage to cloud databases (Firebase / MongoDB).
  - Refactor GUI into a Flask / FastAPI web application.
  - Add link analytics (click counts & geographical metrics).

THANK YOU!